

E12 — Number of Islands (BFS & DFS)

Experiment: 12 | LeetCode: #200 | CO: CO2, CO3 | BT Level: BT5

Problem Statement

Given an $m \times n$ 2D binary grid `grid` which represents a map of '1's (land) and '0's (water), return the number of islands.

An **island** is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are surrounded by water.

Example 1:

Input:

```
grid = [
  ["1","1","1","1","0"],
  ["1","1","0","1","0"],
  ["1","1","0","0","0"],
  ["0","0","0","0","0"]
]
```

Output: 1

Example 2:

Input:

```
grid = [
  ["1","1","0","0","0"],
  ["1","1","0","0","0"],
  ["0","0","1","0","0"],
  ["0","0","0","1","1"]
]
```

Output: 3

Constraints:

- $m == \text{grid.length}, n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 300$
- `grid[i][j]` is '0' or '1'

Concept: Graph View of the Grid

Model the grid as an **implicit graph**:

- Each cell (r, c) with value '1' is a **node**
- Two land cells are **connected** if they are horizontally or vertically adjacent
- An **island** = a **connected component** of land cells

Strategy: Iterate over every cell. When a '1' is found, increment the island count and flood-

fill (BFS or DFS) to mark all connected land cells as visited (so they are not counted again).

Solution 1: DFS (Recursive Flood Fill)

```
def numIslands_dfs(grid):
    """
    DFS flood-fill - marks visited cells by overwriting '1' with '0'.
    Time: O(m * n) | Space: O(m * n) for recursion stack
    """
    if not grid:
        return 0

    rows, cols = len(grid), len(grid[0])
    count = 0

    def dfs(r, c):
        # Base case: out of bounds or water/visited
        if r < 0 or r >= rows or c < 0 or c >= cols or grid[r][c] != '1':
            return
        grid[r][c] = '0' # Mark visited (sink the land)
        dfs(r + 1, c)
        dfs(r - 1, c)
        dfs(r, c + 1)
        dfs(r, c - 1)

    for r in range(rows):
        for c in range(cols):
            if grid[r][c] == '1':
                count += 1
                dfs(r, c) # Flood-fill the entire island

    return count
```

Solution 2: BFS (Queue-based Flood Fill)

```
from collections import deque

def numIslands_bfs(grid):
    """
    BFS flood-fill using a queue.
    Time: O(m * n) | Space: O(min(m, n)) for queue in worst case
    """
    if not grid:
        return 0

    rows, cols = len(grid), len(grid[0])
    count = 0
    directions = [(1, 0), (-1, 0), (0, 1), (0, -1)]

    for r in range(rows):
```

```

for c in range(cols):
    if grid[r][c] == '1':
        count += 1
        grid[r][c] = '0'    # Mark as visited immediately
        queue = deque([(r, c)])

        while queue:
            row, col = queue.popleft()
            for dr, dc in directions:
                nr, nc = row + dr, col + dc
                if 0 <= nr < rows and 0 <= nc < cols and grid[nr][nc] == '1':
                    grid[nr][nc] = '0'    # Mark visited before
                    queue.append((nr, nc))

return count

```

Solution 3: Union-Find (Disjoint Set)

```

class UnionFind:
    def __init__(self, grid):
        rows, cols = len(grid), len(grid[0])
        self.parent = {}
        self.count = 0
        for r in range(rows):
            for c in range(cols):
                if grid[r][c] == '1':
                    self.parent[(r, c)] = (r, c)
                    self.count += 1

    def find(self, x):
        if self.parent[x] != x:
            self.parent[x] = self.find(self.parent[x])    # Path compression
        return self.parent[x]

    def union(self, x, y):
        rx, ry = self.find(x), self.find(y)
        if rx != ry:
            self.parent[rx] = ry
            self.count -= 1

def numIslands_uf(grid):
    """
    Union-Find approach.
    Time: O(m * n * α(m*n)) ≈ O(m * n)    |    Space: O(m * n)
    """
    if not grid:
        return 0

    rows, cols = len(grid), len(grid[0])
    uf = UnionFind(grid)

```

```

for r in range(rows):
    for c in range(cols):
        if grid[r][c] == '1':
            grid[r][c] = '0'
            for dr, dc in [(1,0), (-1,0), (0,1), (0,-1)]:
                nr, nc = r + dr, c + dc
                if 0 <= nr < rows and 0 <= nc < cols and grid[nr][nc] == '1':
                    uf.union((r, c), (nr, nc))

return uf.count

```

Detailed Trace

Example 2: 3 Islands

Grid (initial):

```

1 1 0 0 0
1 1 0 0 0
0 0 1 0 0
0 0 0 1 1

```

BFS Trace:

```

(0,0) = '1' → count=1, enqueue (0,0), mark '0'
Process (0,0): neighbors (1,0)→'1', (0,1)→'1' → enqueue both, mark '0'
Process (1,0): neighbors (0,0)→'0', (2,0)→'0', (1,1)→'1' → enqueue (1,1)
Process (0,1): neighbors (0,0)→'0', (0,2)→'0', (1,1)→'1' → enqueue (1,1)
Process (1,1): no unvisited land neighbors
Queue empty. Island 1 fully visited.

```

Grid now:

```

0 0 0 0 0
0 0 0 0 0
0 0 1 0 0
0 0 0 1 1

```

```

(2,2) = '1' → count=2, enqueue (2,2), mark '0'
Process (2,2): no unvisited land neighbors
Queue empty. Island 2 fully visited.

```

```

(3,3) = '1' → count=3, enqueue (3,3), mark '0'
Process (3,3): neighbor (3,4)→'1' → enqueue (3,4), mark '0'
Process (3,4): no unvisited land neighbors
Queue empty. Island 3 fully visited.

```

```

return 3 ✓

```

Complexity Summary

Approach	Time	Space
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DFS (recursive) $O(m \times n)$ $O(m \times n)$ — recursion stack

BFS (queue) $O(m \times n)$ $O(\min(m, n))$

Union-Find $O(m \times n)$ $O(m \times n)$

Key Takeaways

- **DFS** is the simplest to implement; risk of stack overflow on very large grids.
- **BFS** avoids deep recursion; queue size at most proportional to the perimeter of an island.
- **Union-Find** is useful when the grid is dynamic (cells changing from water to land).
- Marking visited cells in-place ('1' $\rightarrow$ '0') avoids extra memory for a visited set.